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Sri Lanka Institute of Information Technology

B.Sc.(Hons) in Computer Science/ Computer Systems Engineering

Final Examination

Year 2, Semester 1 (2025)

IE2024 – Probability and Statistics

Duration: 2 Hours

May, 2025

Instructions to Candidates:

- This paper is preceded by a 10-minutes reading period. The supervisor will indicate when answering may commence.
- This paper has 4 questions. Answer all questions in the booklet given.
- The total mark for the paper is 100.
- Nonprogrammable calculators are allowed to be used.
- Other electronic devices capable of storing and retrieving text, including mobile phones are not allowed.
- This paper contains 5 pages, including the cover page.

Question 1 (25 Marks)

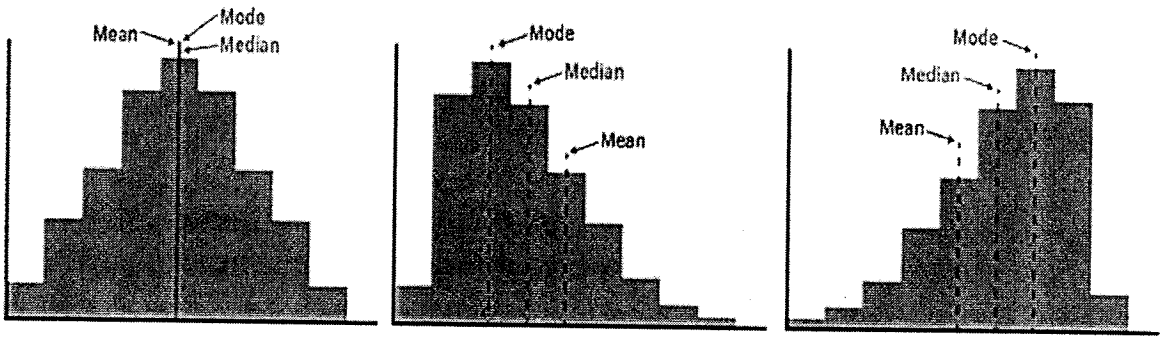
Sampling involves choosing a smaller group of individuals or data points from a larger population to serve as a representation of that population. This allows researchers to draw conclusions about the entire group without having to examine every individual, which conserves both time and resources. After the sample is chosen, descriptive statistics are applied to summarize and highlight the key characteristics of the data.

a) Compare and contrast Probability Sampling and Non-Probability Sampling. (4 Marks)

b) Suppose that a sample of size **n** was taken from a population and the variance of that sample was found as **a**. Then that sample itself was taken as a population for another study and then the variance of that population was found as **b**. Show that, (7 Marks)

$$n = \frac{a}{a - b}$$

c) Critically discuss the behavior of the mean, median and mode of each of the following histogram. (9 Marks)



d) An observatory tracks the energy generated from star observations and the number of new star discoveries each day. To maintain research efficiency, the observatory aims for an energy-to-discovery ratio of 50 joules per new star discovery (Ex- if 100 new stars are discovered in a day, the observatory should generate 5,000 joules of energy that day). Over the last 4 days, the observatory generated 3,800 joules, 4,200 joules, 3,500 joules, and 3,900 joules of energy, while discovering 70, 80, 60, and 70 new stars, respectively. What is the number of new star discoveries the observatory needs on the fifth day if it generates 4,500 joules of energy, in order to achieve a mean energy-to-discovery ratio of 50 joules per star over the 5-day period? (5 Marks)

Question 2 (25 Marks)

Probability theory is a field of mathematics focused on measuring uncertainty and studying random events. It offers a structure for representing and interpreting unpredictable occurrences, allowing us to make predictions using data and probability distributions.

a) Jenny was born between August 26 and 30 (26 and 30 excluding). Her year of birth is also unknown. What is the probability of Jenny being born on a Monday? (5 Marks)

b) Suppose that the error in the reaction temperature, in °C, for a controlled laboratory experiment is a continuous random variable X having the probability density function,

$$f_X(x) = \begin{cases} Cx^2 & -1 < x < 2 \\ 0 & Otherwise \end{cases}$$

i. Find the value of C (3 Marks)

ii. Find P(0<x<1) (3 Marks)

iii. Find the Expected Value (3 Marks)

iv. Find the Variance (3 Marks)

v. Find the Cumulative Density Function (3 Marks)

c) Suppose a researcher is conducting an experiment to study the number of times rats press a lever to receive food. Let's assume that the rate at which the rats press the lever is 4 times per minute on average. Find the probability that a rat presses the lever exactly 10 times during a 4-minute observation period. (5 Marks)

Question 3 (25 Marks)

Statistical inference involves using probability theory and statistical methods to make conclusions about a population based on information from a sample. It helps us estimate parameters, test hypotheses, and make predictions about future outcomes. However, it requires assumptions and may involve sources of uncertainty.

a) Show that for any random variable, (5 Marks)

$$V(X) = E(X^2) - (E(X))^2$$

b) Briefly explain the following conditions of a point estimator.

   a. It should be an unbiased estimator. (2 Marks)

   b. It should be an efficient estimator. (2 Marks)

c) Briefly explain the difference between a point estimator and an interval estimator for a population parameter. (3 Marks)

d) From a class 4 male students and 3 female students were drawn and their mathematics marks were checked. The average mark in the male sample is 72 and the variance is 132. For the female sample the average is 60 and the variance is 100. Assuming the male and female groups are normally distributed and have equal variances, find 95% confidence interval for the difference between the mean of the mathematics marks for male and female. ($t_{2.5\%,5}$=2.57) (5 Marks)

e) The ANOVA test was performed for comparing three population means (A, B, C). There were 5 observations under each sample, taken from each population. Following details are given,

Source of variation	Degrees of freedom	Sum of Squares	Mean Sum of Squares	F- Statistic
Between Groups	P	4,716.4	R	T
Within Groups	Q	1,119.6	S	
Total				

i. Write the Null & Alternative hypotheses for the above scenario. (2 Marks)

ii. Find the values of P, Q, R, S and T (4 Marks)

iii. Discuss the conclusion at 5% level of significance. ($F - table = 3.89$) (2 Marks)

Question 4 (25 Marks)

Linear regression is a simple and widely used statistical method for understanding the relationship between two variables. It helps us find the best-fitting straight line through a set of data points, where one variable (the independent variable) is used to predict the other (the dependent variable). The goal of linear regression is to identify a line that minimizes the distance between the actual data points and the line itself. This line represents the general trend of the data, and we can use it to make predictions or understand how changes in one variable are associated with changes in the other.

a) Mention three assumptions in Linear Regression and briefly explain. (6 Marks)

b) Suppose there is a categorical independent variable **X** which has three categories **a, b, c**. Discuss the process of making dummy variables for this categorical variable? (4 Marks)

c) Assume for a linear regression model,

$$\text{Total Sum of Squares (TSS)} = 1000$$

$$\text{Sum of Squares of Error (SSE)} = 200$$

Find the Coefficient of Determination for this model using the above values. (5 Marks)

d) Evaluate the statement "**Correlation does not imply causation**". (4 Marks)

e) Assume an estimated simple linear regression model for predicting the GPA with the Study Hours is as below.

$$\widehat{GPA} = 0.2 + 0.5 \times \widehat{Study_Hours}$$

i. Interpret the intercept (0.2) of the model (3 Marks)

ii. Interpret the slope (0.5) of the model (3 Marks)

----------End of the Question Paper ----------